Title

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Abstract

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1 Introduction

Viral kinetics refers to the changes in viral levels within an infected individual over time. Viral kinetics (VK) models are within-host mathematical models that describe these dynamics. VK modeling provides investigators with curves of viral load over the course of an infection as well as estimates of key epidemiological parameters.

The original VK models were developed for the study of HIV in the 1990s [1]. Since then, VK models have proved useful for a variety of tasks. VK modeling can be used to address clinical questions, concerning topics like drug resistance [2] and antiviral therapies [3], as well as epidemiological questions, such as understanding trasmissibility over time [CITE] and determining optimal surveillance strategies [4]. Modeling viral kinetics also allows us to understand how infections with the same virus might vary depending factors such as the variant [5], the vaccination status of the host [6], whether the host has been infected before [7], and demographic characteristics of the host, such as age [5].

There are two main categories of VK models: mechanistic and phenomonological.

The fundamental challenge of building VK models is the amount and type of data needed, and the cost associated with collecting it. If too few samples are collected, the model will struggle to accurately estimate the dynamics of viral concentration, producing poor estimates of epidemiological parameters. Collecting high numbers of samples presents its own challenges, increasing discomfort for subjects and incurring high costs for the investigator [8]. Data for fitting VK models may be collected via a challenge study or from naturally occurring infections in some population. Popular approaches for collecting naturally-occurring infection data, such as serial sampling after an initial positive and aggregating single samples from individuals, produce models with fundamental flaws [5]. Gold-standard data for VK modeling emerged as a result of surveillance efforts during the SARS-CoV-2 pandemic, allowing for work which demonstrated the value of building VK models and of collecting the data necessary to fit them [CITE].

2 Study Design and Optimized Assay Approaches

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We develop methods for fitting a viral kinetics model given a set of (is there a word more general than "saliva" that I can use here?) samples collected from M individuals over N days. These samples can be arranged into an M by N matrix $\mathbf{A}$, where an entry $A_{i,j}$ is the sample collected from participant i on day j. We say that $A_{i,j} > 0$ if individual i had viral RNA concentration above the limit of detection (LOD) of our assay on day j. Otherwise, $A_{i,j} = 0$. Observing the value of an entry in the matrix amounts to assaying the corresponding sample.

Data of this kind might emerge in a variety of ways. Samples originally collected for a study focusing on one pathogen can be arranged into this format and used to fit a model of another pathogen. (Need another example of where this data might come from.) Of course, researchers might collect samples from a population with the express purpose of employing the assaying approaches proposed here.

The ultimate goal of this work is to develop efficient algorithms to obtaining a fit model of the viral kinetics of a pathogen, starting from the matrix $\mathbf{A}$ whose values we have not yet observed. There are two main types of algorithms we propose towards this end: the first locates infections within the sample matrix and the second fits a viral kinetics model to those infections.

Unless researchers specifically target individuals who they know are likely to become infected within a fixed period of time (an approach which incurs its own high expenses), $\mathbf{A}$ will be a sparse matrix. If we were to assay every sample in $\mathbf{A}$, most of the results will be negative, offering little information to our model. That is, only a small proportion of the assays perfomed in the complete assaying case are informative. This suggests that we can locate the infections within $\mathbf{A}$ using far fewer than the $M * N$ assays required by the complete testing approach. But how many tests are actually needed, and how should we distribute them among the samples?

2.1 Infection Discovery

The problem of infection discovery amounts to identifying continuous sequences of positive entries within the rows of $\mathbf{A}$. We need not record the entire infection from start to finish, but rather identify at least one of the positive values within it. This can be achieved by the following algorithm.

Suppose you are given the matrix $\mathbf{A}$ with unobserved entries and cumulative incidence c. Assume that the duration of each infection ℓ_i is drawn from some continuous positive distribution with practical lower bound ℓ_0 such that $\mathbb{P}(\ell_i < \ell_0) < \alpha$ for some small constant α. (how are we doing this in sims since we're working in discrete time?)

The algorithm proceeds as follows. (a diagram might be useful for this) Within each row: pool the samples and apply a single assay to the pool (gotta show that we'll stay above LO). For each row i on which the assay returns a negative result, set $A_{i,j} = 0$ for all j. This row does not contain an infection and can now be removed from consideration. For each row i on which the assay returned a positive result, individually assay entries at an interval of ℓ_0. That is, assay samples $A_{i,0}, A_{i,\ell_0}, A_{i,2\ell_0}, A_{i,3\ell_0}, \ldots$.

Both the accuracy of this approach and the cost reduction it offers can be expressed in closed form. We quantify the accuracy of the infection discovery algorithm by the probability that an arbitrary trajectory is not missed. An infection is missed if none of the samples taking during its duration are assayed by the algorithm. The probability that an arbitrary infection i is missed by the infection discovery algorithm is described by the inequality

$$\mathbb{P}(\text{miss } i) < \alpha - \frac{\alpha}{\ell_0}\mathbb{E}\left[\ell_i | \ell_i < \ell_0\right] \tag{1}$$

The cost C of running the infection discovery algorithm over matrix $\mathbf{A}$ is

$$C = MN(\frac{1}{N} + \frac{c}{\ell_0}). \tag{2}$$

This reduces the cost of the brute force approach by a factor of $(\frac{1}{N} + \frac{c}{\ell_0})$. Infection discovery gets cheaper as the number of individuals sampled increases and the ratio of the cumulative incidence to the sampling interval decreases. Derivations for Equations 1 and 2 can be found in Section 3.1.

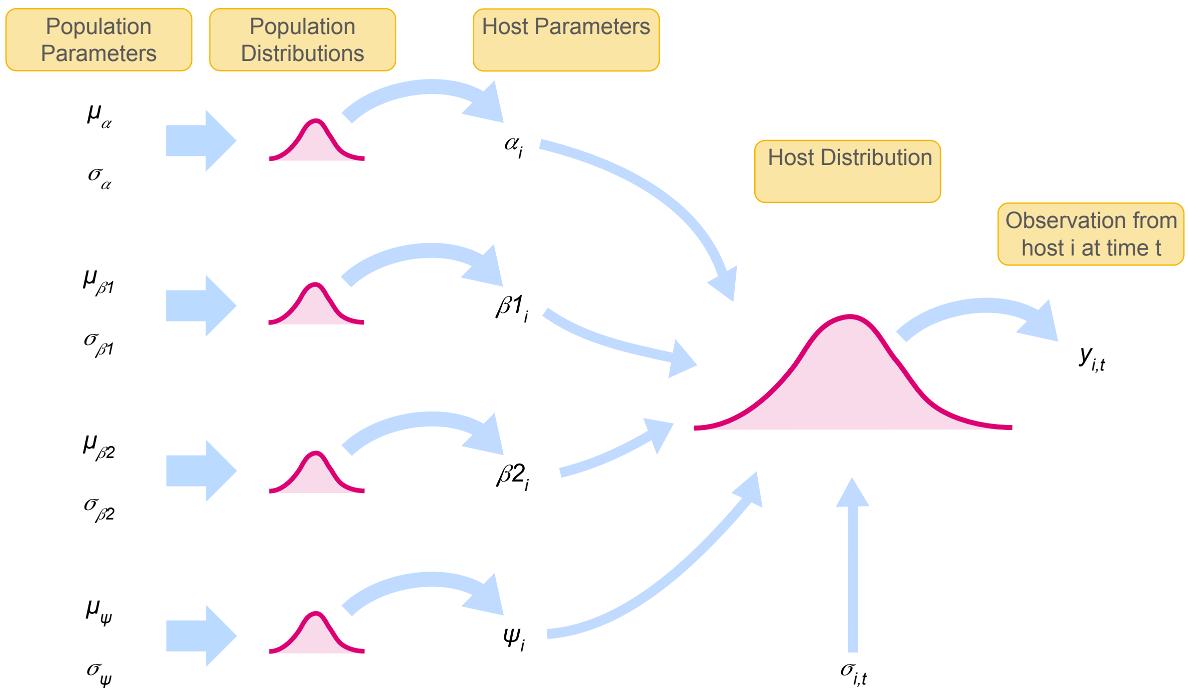

Figure 1: The Bayesian hierarchical model.

3 Appendix

3.1 Accuracy and Cost of Infection Discovery

For an arbitrary infection i contained in sample matrix $\mathbf{A}$, the probability the infection discovery algorithm misses i can be expressed as follows.

$$
\begin{aligned}
\mathbb{P}(\text{miss } i) &= \mathbb{P}(\ell_i \geq \ell_0)\mathbb{P}(\text{miss } i|\ell_i \geq \ell_0) + \mathbb{P}(\ell_i < \ell_0)\mathbb{P}(\text{miss } i|\ell_i < \ell_0) \\
&= \mathbb{P}(\ell_i < \ell_0)\mathbb{P}(\text{miss } i|\ell_i < \ell_0) \\
&= \int_0^{\ell_0} \mathbb{P}(\ell_i)\mathbb{P}(\text{miss } i|\ell_i) d\ell_i \\
&= \int_0^{\ell_0} \mathbb{P}(\ell_i) \left(1 - \frac{\ell_i}{\ell_0}\right) d\ell_i \\
&= \int_0^{\ell_0} \mathbb{P}(\ell_i) - \int_0^{\ell_0} \mathbb{P}(\ell_i)\frac{\ell_i}{\ell_0} d\ell_i \\
&= \mathbb{P}(\ell_i < \ell_0) - \frac{\mathbb{P}(\ell_i < \ell_0)}{\ell_0}\mathbb{E}\left[\ell_i|\ell_i < \ell_0\right] \\
&< \alpha - \frac{\alpha}{\ell_0}\mathbb{E}\left[\ell_i|\ell_i < \ell_0\right]
\end{aligned}
$$

We now consider the cost of performing infection discovery on $\mathbf{A}$. Suppose each assay costs σ. Then the cost of the pooling step of the infection discovery algorithm is σM. The cost of the interval sampling step is

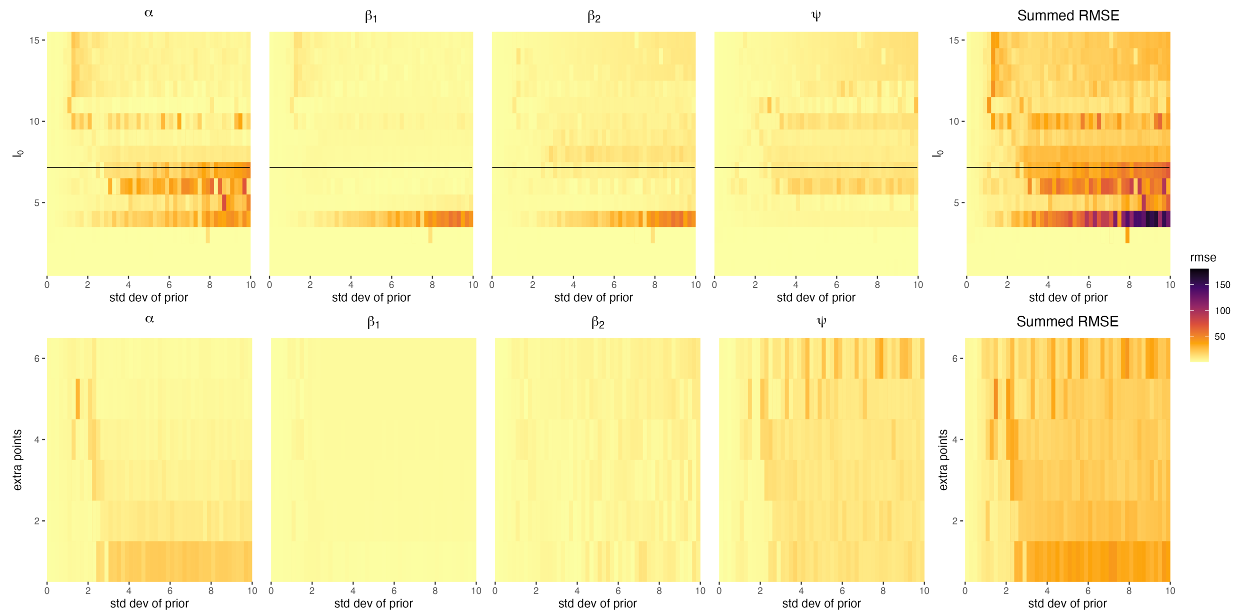

Figure 2: Parameter sweeps.

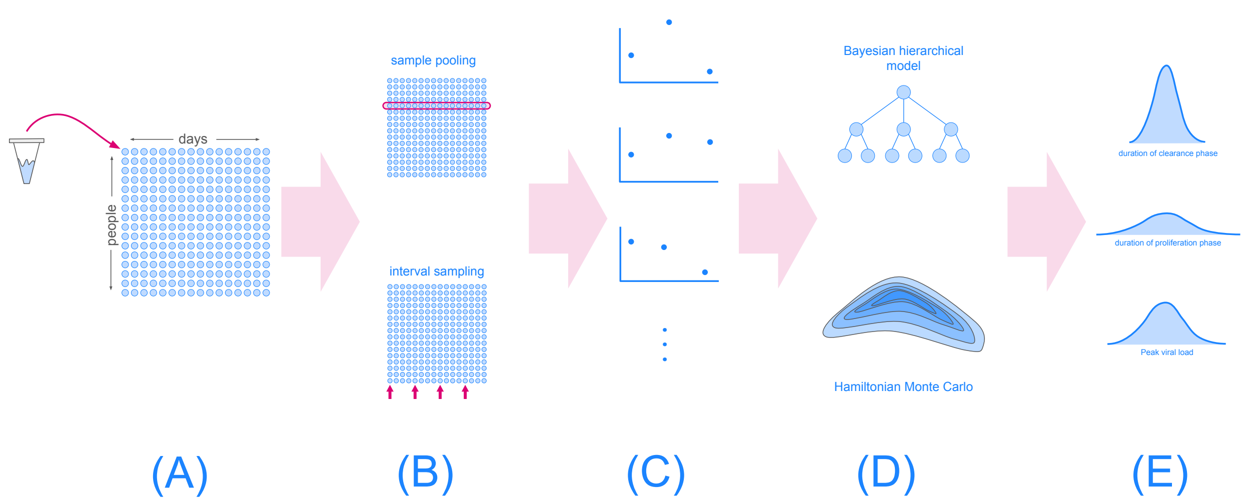

Figure 3: The model fitting pipeline. In Phase (A), subjects provide saliva samples over several days to create a matrix of samples with dimensions subjects by days. In Phase (B), sample pooling and interval sampling are used to identify the locations of infections in the matrix. This results in a series of partially sampled viral load trajectories. Using Hamiltonian Monte Carlo, the Bayesian hierarchical model specified in 1 is fit to the partially sampled trajectories. The fit model provides probability distributions over possible values of the epidemiological parameters of the pathogen.

References

[1] Alan S Perelson, Avidan U Neumann, Martin Markowitz, John M Leonard, and David D Ho. Hiv-1 dynamics in vivo: virion clearance rate, infected cell life-span, and viral generation time. *Science*, 271 (5255):1582–1586, 1996.

[2] Libin Rong, Ruy M Ribeiro, and Alan S Perelson. Modeling quasispecies and drug resistance in hepatitis c patients treated with a protease inhibitor. *Bulletin of mathematical biology*, 74(8):1789–1817, 2012.

[3] Frank S Heldt, Timo Frensing, Antje Pflugmacher, Robin Gröpler, Britta Peschel, and Udo Reichl. Multiscale modeling of influenza a virus infection supports the development of direct-acting antivirals. *PLoS computational biology*, 9(11):e1003372, 2013.

[4] Stephen M Kissler, Joseph R Fauver, Christina Mack, Scott W Olesen, Caroline Tai, Kristin Y Shiue, Chaney C Kalinich, Sarah Jednak, Isabel M Ott, Chantal BF Vogels, et al. Viral dynamics of acute sars-cov-2 infection and applications to diagnostic and public health strategies. *PLoS biology*, 19(7): e3001333, 2021.

[5] Timothy W Russell, Hermaleigh Townsley, Sam Abbott, Joel Hellewell, Edward J Carr, Lloyd AC Chapman, Rachael Pung, Billy J Quilty, David Hodgson, Ashley S Fowler, et al. Combined analyses of within-host sars-cov-2 viral kinetics and information on past exposures to the virus in a human cohort identifies intrinsic differences of omicron and delta variants. *PLoS Biology*, 22(1):e3002463, 2024.

[6] Ruian Ke, Pamela P Martinez, Rebecca L Smith, Laura L Gibson, Chad J Achenbach, Sally McFall, Chao Qi, Joshua Jacob, Etienne Dembele, Camille Bundy, et al. Longitudinal analysis of sars-cov-2 vaccine breakthrough infections reveals limited infectious virus shedding and restricted tissue distribution. In *Open Forum Infectious Diseases*, volume 9, page ofac192. Oxford University Press, 2022.

[7] Stephen M Kissler, James A Hay, Joseph R Fauver, Christina Mack, Caroline G Tai, Deverick J Anderson, David D Ho, Nathan D Grubaugh, and Yonatan H Grad. Viral kinetics of sequential sars-cov-2 infections. *Nature Communications*, 14(1):6206, 2023.

[8] Laetitia Canini and Fabrice Carrat. Viral kinetics studies on influenza: when and how many times are nasal samples to be collected? *Influenza and Other Respiratory Viruses*, 5:144–147, 2011.